| Mycorrhizae | | |
|---|---|---|
| **Ectomycorrhizae** | **Endomycorrhizae** | **Ectendomycorrhizae** |
| The fungal mycelium forms a dense sheath around the root called mantle. The hyphal network penetrate the intercellular spaces of the epidermis and cortex to form Hartignet. Example: *Pisolithus tinctorius* | The hyphae grows mainly inside the roots, penetrate the outer cortical cells of the plant root. A small portion of the mycelium is found outside the root. This form is also called Vesicular Arbuscular Mycorrhizal fungi (VAM Fungi) due to the presence of Vesicle or arbuscle like haustoria<br>1. Arbuscular mycorrhizae(VAM) Example: *Gigaspora*<br>2. Ericoid mycorrhizae -Example: *Oidiodendron*<br>3. Orchid mycorrhizae -Example: *Rhizoctonia* | The fungi form both mantle and also penetrates the cortical cells. |

and water to the plants.

- Provides drought resistance to the plants
- Protects roots of higher plants from the attack of plant pathogens

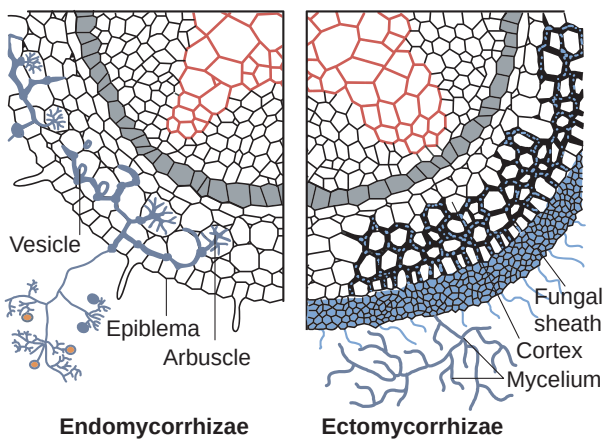



**Figure 1.31:** T.S. of root showing mycorrhizae

### 1.5.10 Lichens

The symbiotic association between algae and fungi is called lichens. The algal partner is called Phycobiont or Photobiont., and the fungal partner is called Mycobiont. Algae provide nutrition for fungal partner in turn fungi provide protection and also help to fix the thallus to the substratum through rhizinae. Asexual reproduction takes place through fragmentation, Soredia and Isidia. Phycobionts reproduce by akinetes, hormogonia, aplanospore etc., Mycobionts undergo sexual reproduction and produce ascocarps.

**Classification**

- Based on the habitat lichens are classified into following types: **Corticolous**( on Bark) **Lignicolous**(on Wood) **Saxicolous**(on rocks) **Terricolous**(on ground) Marine(on siliceous rocks of sea) Fresh water(on siliceous rock of fresh water).
- On the basis of morphology of the thallus they are divided into **Leprose** (a distinct fungal layer is absent) **Crustose**-crust like; Foliose-leaf like; **Fruticose**- branched pendulous shrub like (Figure 1.32).
- The distribution of algal cells distinguishes lichens into two forms namely **Homoiomerous** (Algal cells